

Speech-To-Text Translation Using End-To-End Transformer Models

Traducción de Voz a Texto Mediante Modelos Transformadores Extremo a Extremo

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Abstract

Automated speech-to-text translation is a crucial task in natural language processing, facilitating real-time communication across languages in applications such as video subtitling and multilingual accessibility services. Traditional systems often employ a cascade approach, where an automatic speech recognition (ASR) model first transcribes audio into text, followed by a machine translation (MT) model that translates the text into the target language. However, cascade models have an important challenge: transcription errors from the ASR model are propagated to the MT model, negatively impacting the translation quality. This study proposes an end-to-end translation model based on a transformer architecture, which integrates ASR and MT into a single unified structure. This approach reduces error propagation and improves translation accuracy. In the experiments carried out, the CoVoST2 dataset was used with the Whisper-Large-V3 model, the direct translation achieved a BLEU score of 41.85, significantly outperforming the cascaded translation approach, which scored 31.13 BLEU. These results clearly demonstrate the effectiveness and improvement of the proposed method.

Keywords: Speech-to-Text Translation, Automatic Speech Recognition, Machine Translation, Transformer, End-to-End Model

Resumen

La traducción automática de voz a texto es esencial en el procesamiento del lenguaje natural, ya que permite la comunicación en tiempo real entre idiomas, siendo útil en subtitulación y servicios de accesibilidad multilingües. Tradicionalmente, se ha utilizado un enfoque en cascada: primero, un modelo de reconocimiento automático de voz (ASR) transcribe el audio, y luego, un modelo de traducción (MT) convierte el texto al idioma objetivo. No obstante, este enfoque tiene un problema crítico: los errores en la transcripción afectan directamente la calidad de la traducción. Para mitigar este inconveniente, este estudio propone un modelo de traducción de extremo a extremo basado en la ar-

quitectura transformador, que combina ASR y MT en una sola estructura. Este modelo unificado reduce la propagación de errores y mejora la precisión de las traducciones. En los experimentos realizados, se utilizó el conjunto de datos CoVoST2 con el modelo Whisper-Large-V3, la traducción directa logró un resultado de 41.85 BLEU, superando significativamente a la traducción en cascada, que obtuvo 31.13 BLEU, lo que evidencia una mejora considerable con el enfoque propuesto.

Palabras claves: Traducción de voz a texto, Reconocimiento automático de voz, Traducción automática, Transformador, Modelo de extremo a extremo

1 Introduction

Automatic speech recognition (ASR) and machine translation (MT) are closely related in the field of artificial intelligence [1, 2], with applications in various domains such as automatic subtitling of audiovisual content, multilingual virtual assistants, real-time simultaneous interpretation, and accessibility in global customer service [3]. The combination of ASR and MT enables systems to translate what it says into other languages immediately. In applications such as multilingual video conferencing, online education, and automatic subtitling, integrating these technologies can greatly enhance communication between people speaking different languages, promoting global collaboration and accessibility. Traditionally, these tasks have been approached through cascade models, where ASR first transcribes the audio into text, and then the MT model translates the resulting text into another language. However, cascade systems exhibit an inherent weakness: error propagation between stages [3, 4, 5]. Mistakes made during transcription are carried over to the translation stage without correction, impacting the final output quality.

To mitigate such issues, various researchers have been proposing end-to-end (E2E) models [6, 7, 8, 9]. Which eliminate the need for an intermediate step between ASR and MT, allowing the system to process audio directly and generate the final translation without

relying on a prior transcription. This approach offers several advantages, as it reduces the error accumulation that occurs in cascade systems and enables the model to leverage audio and textual information more holistically. In addition, end-to-end models, particularly those based on transformer architectures [10, 11], have shown highly effective in simultaneous transcription and translation tasks [12, 13, 14], improving the accuracy and fluency of translations even in acoustically challenging conditions. Thus, OpenAI’s Whisper model [15] is a recent example of these advancements, standing out for its ability to handle multilingual tasks and robust recognition in a single, continuous process. For that reason, this paper proposes Whisper as an end-to-end solution to address the error propagation problem in ASR and MT cascade systems based on Transformer [16]. It can handle large volumes of multilingual data and speech in diverse acoustic conditions, making it a more robust and efficient approach to audio transcription and translation than traditional cascade systems.

2 Related works

In recent developments, Wang et al. introduced FAIRSEQ S2T [17] as an extension of the FAIRSEQ framework for speech-to-text (S2T) modeling tasks such as end-to-end speech recognition and speech-to-text translation. This framework was built with a strong focus on scalability and extensibility based on RNN, Transformer, and Conformer architectures.

Likewise, Le et al. developed ComSL [4], a pre-trained speech-language model, which addresses challenges such as high data and GPU requirements and the modality gap between speech and text; integrating multimodal learning that optimizes the efficiency of spoken language tasks. Achieving a new state-of-the-art average BLEU score of 31.5 on the multilingual speech-to-English task across 21 languages, using the CoVoST2 dataset.

Similarly, Hu et al. proposed GenTranslate [18], a novel generative approach leveraging large language models (LLMs) to enhance multilingual speech and machine translation. Unlike traditional methods that rely on beam search decoding and single-hypothesis selection, GenTranslate integrates multiple translation candidates from the N-best list using the linguistic reasoning capabilities of LLMs, leading to higher-quality outputs. To further improve performance, the Hypo-Translate dataset, containing over 592K hypothesis-translation pairs across 11 languages, was introduced for fine-tuning.

Furthermore, Communication et al. introduced SeamlessM4T [19], a massively multilingual and multimodal machine translation model supporting various translation tasks, including speech-to-speech, speech-to-text, text-to-speech, text-to-text, and automatic speech recognition (ASR) across 100 languages. It

is the first framework capable of bidirectional translation between English and multiple languages for both speech and text. On the Fleurs benchmark, SeamlessM4T achieves a 20% BLEU improvement over the previous state-of-the-art in direct speech-to-text translation, surpassing cascaded models by 1.3 BLEU points in speech-to-text and 2.6 ASR-BLEU points in speech-to-speech translation.

On the other hand, Jia et al. introduced CVSS [20], a massively multilingual speech-to-speech translation (S2ST) corpus, demonstrating the advantages of end-to-end direct models over traditional cascade systems. Cascade S2ST models trained on CVSS surpassed previous state-of-the-art results on CoVoST 2, improving text translation by 5.8 BLEU points. However, direct models trained from scratch on CVSS achieved comparable performance, with only 0.1 to 0.7 BLEU differences, especially when initialized from pre-trained Speech-to-Text (ST) models.

Furthermore, Chen et al. introduced Maestro [21], a self-supervised modality matching algorithm designed to unify speech and text representations without explicit modality conversion, addressing challenges such as interference between modalities and high computational complexity. By leveraging sequence alignment, duration prediction, and masked language modeling, Maestro enhances multilingual ASR and speech translation, achieving a state-of-the-art 8% relative WER reduction on VoxPopuli and a 2.8 BLEU improvement on CoVoST-2 across 21 languages.

Additionally, Bapna et al. introduced mSLAM [14], a multilingual speech-language model that unifies speech and text representations through joint pre-training, addressing challenges such as cross-modal alignment and data scarcity in low-resource languages. By integrating w2v-BERT for speech, SpanBERT for text, and CTC losses, mSLAM enables zero-shot text translation and improves speech translation, intent classification, and language identification.

Some research in the review of related works focuses on speech-to-text translation using either cascade models or advanced multilingual frameworks, each varying in accuracy depending on the architecture and training techniques employed. Given that cascade models often propagate errors between the ASR and MT stages. And end-to-end models have shown promising yet underexplored potential, this research uses the Whisper-Large-v3 model as an end-to-end approach to obtaining results that outperform traditional cascade systems in multilingual speech-to-text translation tasks.

3 Classical Method

The traditional ST translation methods involve a cascade approach, employing two independent models: an automatic speech recognition (ASR) system followed by a machine translation (MT) system.

The graphical representation of a fine-tuning for cascaded translation is shown in Figure 1. It will be compared with the proposed end-to-end translation. For that reason, in this model, was used Whisper-Large-V3 as ASR and mBART-Large-50 [22, 23] as MT, as a transformer-based model, which is employed to convert speech into text.

The transcribed output was fed into mBART, a multilingual neural translation model.

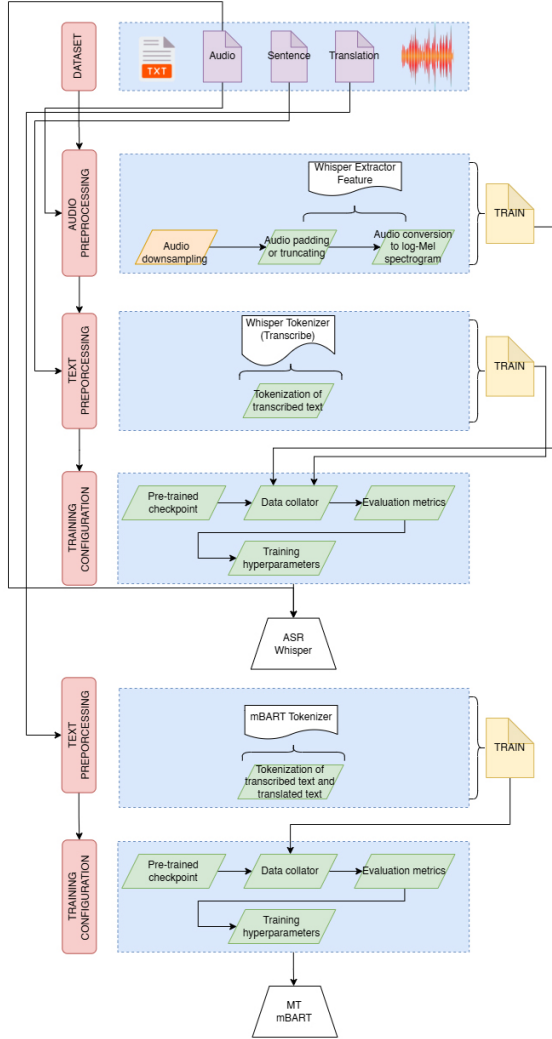


Figure 1: Graphical representation of a fine-tuning for Cascaded translation using Whisper (ASR) and mBART (MT).

4 Proposed Method

Figure 2 shows a graphical representation of the proposed method for fine-tuning in end-to-end translation using Whisper based on Transformer.

4.1 Dataset

The experiments of the proposed method were carried out using the CoVoST2 dataset. The relevant content

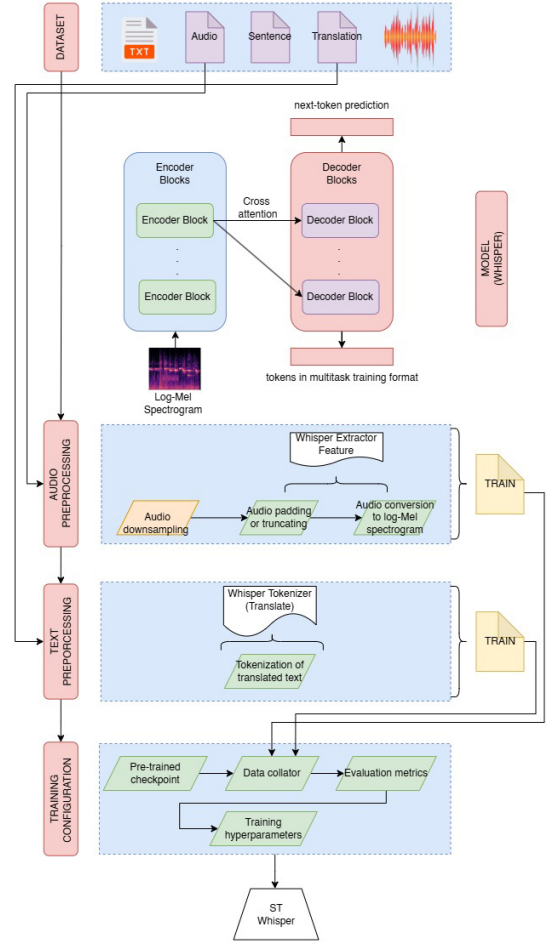


Figure 2: Graphical representation of the proposed method.

of this dataset is the audio language source, the transcription of the audio file in the source language, and the transcription of the audio file in the target language. CoVoST2 [24] is a multilingual corpus that includes over 2000 hours of audio in over 20 languages, along with their corresponding aligned transcriptions and translations. We used Spanish-to-English CoVost2 that relies on another dataset, Mozilla Common Voice 4 [25], which specifically provides Spanish Audio, Spanish Text, and English Translated Text.

- **Spanish Audio:** Recorded speech fragments, taken from the Common Voice initiative, with many accents and pronunciation styles. These audios are segmented at the sentence level and generally have an average length of 5-10 seconds per fragment, or can vary slightly depending on the speech content and the original segmentation in Common Voice.
- **Spanish Text:** Accurate transcriptions of the spoken content, faithfully reflecting the speech in the source language and useful for automatic speech recognition tasks.

- English Translated Text: Translations aligned with the Spanish transcripts, designed to serve as a reference in automatic speech-to-text translation tasks.

CoVoST2 includes a few empty audio files (0 bytes), from Common Voice version 4. We excluded these audios.

4.2 Transformer Model

4.2.1 Speech Translation Task

This process translates input speech " U " in a target language. Optionally, this task includes an intermediate transcription " X " of the source language. Formally, given a dataset D composed of triples (u_i, x_i, v_i) where u_i represents the speech features, x_i is the transcription in the source language, and v_i is the translated text, the goal is to minimize the conditional probability $P(v|u; \theta)$, which is decomposed using the autoregressive conditional probability principle [3]. See equation 1.

$$p(v|u; \theta) = \prod_{t=1}^{T_v} p(v_t | v_{<t}, u; \theta) \quad (1)$$

Where T_v : Represents the length of the translated text sequence, v_t : Is the token at position t in the translated sequence, $v_{<t}$: Is the set of tokens generated prioritizing position t , u : Refers to the input speech features and θ : Denotes the model parameters.

In essence, it indicates that the probability of a complete translated sequence v can be calculated as the product of the conditional probabilities of each token v_t , given the context of the previous tokens $v_{t<}$ and the input speech features u .

The model is trained by minimizing the negative log-likelihood over the dataset D , expressed in equation 2 [3].

$$l(\theta|D) = - \sum_{i=1}^n \log P(v^i | u^i; \theta) \quad (2)$$

Where n is the number of data pairs in the training set, and $P(v^i | u^i; \theta)$ is the conditional probability of the translated text v^i given the input speech u^i . This loss function is optimized using an encoder-decoder architecture with attention mechanisms. The encoder maps the input speech u into a hidden representation h , while the decoder generates the tokens of v one by one, based on the representation h , the previously generated tokens, and an attention vector α [16].

4.2.2 Encoder to decoder Model - Whisper

It is an encoder-decoder model [15], developed by OpenAI, based on the transformer architecture [16], also known as a sequence-to-sequence or end-to-end model. It utilizes audio signal encoding based on spectrograms, which feed the transformer encoder-decoder.

The encoder processes the audio and extracts high-level representations, while the decoder generates the corresponding text sequences. The Whisper models were trained with 680,000 hours of multilingual audio of varying quality, to ensure generalization to different types of audio input. The models were trained on either English-only data on the task of speech recognition or multilingual data on speech recognition and speech translation. For speech recognition, the model predicts transcriptions in the same language as the audio. For speech translation to English. For this reason, in this research was used Whisper in the end-to-end translation (ST) and cascade translation (ASR) stages. For the implementation of the systems (Cascaded and End-to-End), the transformers library from Hugging Face [26] was used, allowing access to pre-trained models such as Whisper, specifically large-V3, for ASR and ST, as well as mBART, specifically large-50, for translation in the cascade approach. Additionally, the preprocessing tools available in the same library were utilized, facilitating tokenization, text normalization, and the conversion from audio into the required representations for each model.

4.3 Audio Pre-Processing

The sampling rate of the audio inputs must match the expected sampling rate of the model. Whisper models require audio inputs with a sampling rate of 16kHz [15], so it is necessary to adjust the inputs accordingly.

4.3.1 Audio Padding Or Truncating

Whisper feature extractor was used, available in Hugging Face [26] to convert audio to the required representations for our model.

The Whisper feature extractor ensures that all audio samples in a batch have a fixed input length of 30 seconds by either padding or truncating them. If a sample is shorter than 30 seconds, it is padded with zeros, which represent silence or the absence of sound. Conversely, if a sample exceeds 30 seconds, it is truncated to fit the required length. Hence an attention mask is not needed when passing audio inputs to the Whisper model as in other audio models. Instead, Whisper has been trained to function without an attention mask, relying on speech signals to determine which parts of the input should be disregarded.

4.3.2 Audio Conversion To Log-Mel Spectrogram

The Whisper feature extractor's second operation was converting the padded audio arrays to log-Mel spectrograms. These spectrograms visually represent a signal's frequencies rather like a Fourier transform. The log-Mel spectrogram is the form of input expected by the Whisper model. The Mel channels (frequency bins) are standard in speech processing and have been

chosen to approximate the human auditory range. Figure 3 shows the log-mel spectrogram using 128 Mel frequency bins of a voice audio.

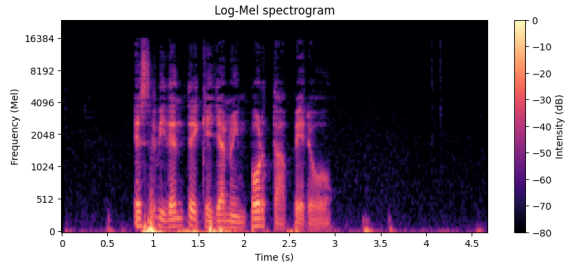


Figure 3: Log-Mel spectrogram of the phrase: "Es evidente que ya no importa ¿pero?"

4.4 Text Pre-Processing

4.4.1 Text normalization

Text normalization is not standardized, making the comparison of results between different works difficult [20]. We did not normalize the text for both training and testing. Table 1 shows an example.

Table 1: Example of normalization (English translation)

Original text	Transformed text
Once you hit your double, you will be "a killer" for the rest of the game.	once you hit your double you will be a killer for the rest of the game

4.4.2 Tokenization

A Whisper tokenizer generates the text output based on the processed audio input. The Whisper model produces text tokens that correspond to the index of the predicted text within the predefined vocabulary dictionary. Like the Whisper feature extractor, we used the Whisper tokenizer available in Hugging Face [26] to convert text to the required representations for our model. The tokenizer is loaded from the pre-trained checkpoint to be ready for our text data. We specified the target language and the task, such as transcription or translation. Table 2 shows an example of translation tokenization.

4.5 Training Configuration

4.5.1 Pre-Trained Checkpoint

A pre-trained checkpoint is loaded to begin the fine-tuning process. There are five different configurations, the four smaller versions come in English-only and multilingual variants, while the larger checkpoints are

Table 2: Example of tokenization for translation from Spanish to English.

Translation
<—startoftranscript—><—en—><—translate—> <—notimestamps—>"Una vez haya alcanzado su doble, será un "asesino" durante el resto del juego."<—endoftext—>

exclusively multilingual. We used the largest model, Whisper-Large-V3, to try to achieve the best possible result. Table 4 shows the details of the Whisper model family.

Table 3: Architecture details of the Whisper model family.

Size	Layers	Width	Heads	Parameters	English-only	Multilingual
tiny	4	384	6	39 M	✓	✓
base	6	512	8	74 M	✓	✓
small	12	768	12	244 M	✓	✓
medium	24	1024	16	769 M	✓	✓
large	32	1280	20	1550 M	X	✓
large-v2	32	1280	20	1550 M	X	✓
large-v3	32	1280	20	1550 M	X	✓

4.5.2 Data Collator

This module is unique and prepares audio and text data for training the Whisper model, ensuring that the data is padded, formatted correctly, and ready to be fed into the model. Hence, the input feature is handled by the feature extractor and the labels by the tokenizer.

4.5.3 Evaluation Metrics

To evaluate the quality of translation in our test data, we used the BLEU (Bilingual Evaluation Understudy) metric [27] from the sacrebleu package. This metric measures translation quality, a commonly used metric in machine translation that compares the similarity between the generated translation and one or more reference translations. BLEU is based on n-gram (word group) matches between the generated translation and the reference, with a higher BLEU score indicating a more accurate and fluent translation. We did not perform any normalization to the text before computing BLEU.

Additionally, the WER (Word Error Rate) metric was used to evaluate the quality of speech recognition as a standard measure of the accuracy of automatic speech recognition (ASR) systems, which represents

the proportion of words incorrectly recognized by the system compared to a reference transcript.

4.5.4 Training Hyperparameters

Table 4 shows all the parameters related to training used in this research.

Table 4: Whisper training hyperparameters.

Hyperparameter	Value
Learning Rate	1e-05
Train batch	16
Eval batch	16
Optimizer	Adam
β	0.95
β_2	0.95
ϵ	1e-05
Learning scheduler type	Linear
Weight decay	0.01
Training epoch	1
Mix Precision Training	Native AMP

4.6 Validation of Results

The test split of the CoVoST 2 corpus was used to evaluate and compare the performance of end-to-end and cascade models in speech-to-text translation.

4.6.1 End-to-end translation

The Whisper-Large-V3 model was used to perform ASR and MT directly from audio to translated text. This approach eliminates the need for an intermediate transcription step. Hence, each audio file in the CoVoST2 test split was fed directly into the Whisper model, which generated the translation into the target language in an end-to-end manner.

4.6.2 Cascaded translation

In this approach, Whisper-Large-V3 was used only to transcribe speech to text in the source language. Subsequently, the Whisper output was processed by the mBART-Large-50 model, a multilingual machine translation model, to generate the translation into the target language.

Whisper first transcribed each audio file in the CoVoST2 test split, and the generated transcriptions were passed into mBART to perform the translation.

5 Experiments

To demonstrate that end-to-end models can achieve better results than cascade models in speech-to-text translation, the results obtained using both approaches were compared. A cascade system combined ASR

using the Whisper-Large-V3 model with MT through the mBART-Large-50 model (Figure 5.2), and an end-to-end system used only the Whisper-Large-V3 model to perform ASR and MT in a single task (Figure 5.2).

In the experiments, the data set mentioned in Section 3.2 was used to evaluate the performance of the proposed Whisper model for Spanish to English translation.

5.1 Direct Translation Baselines

The train split was utilized to fine-tune the Whisper-Large-V3 model in the translation task. We followed the hyperparameters from Table 4. The test split was used to evaluate the speech-to-text translation task, whose performance is reported in Table 6.

5.2 Cascade Translation Baselines

To construct cascade translation baselines, we trained an ASR model on the original CoVoST 2 and connected it to the MT model used to build cascade translation. The training split was used to fine-tune the Whisper-Large-V3 model for the transcription task and the mBART-Large-50 model for the translation task. For both models, the hyperparameters listed in Table 4 were taken into account. The test split was used for the evaluation, and the performance of the ASR task and the MT task was reported in Table 5. Once fine-tuned, we used the test data set again, along with Whisper (ASR) and mBART (MT), to evaluate the speech-to-text translation task, and we report the performance in Table 6.

Table 5: Automatic Speech Recognition (Spanish) and machine translation (Spanish to English) results on CoVoST-2 test sets in terms of BLEU score (MT) and WER score (ASR).

ASR	Es
Whisper-Large-V3	9.364048
MT	Es -> En
mBART-Large-50	43.299020

Table 6: Speech translation results on CoVoST-2 Es->En test sets in terms of BLEU score.

X -> En	Es
Cascaded ASR+MT Methods	
Whisper-Large-V3 + mBART-Large-50	31.13
End-to-end ST Methods	
Whisper-Large-v3	41.8539

6 Analysis

Tables 7 and 8 show a comparison between the results in both direct and cascade translation using state-of-the-art models and the proposed Whisper model.

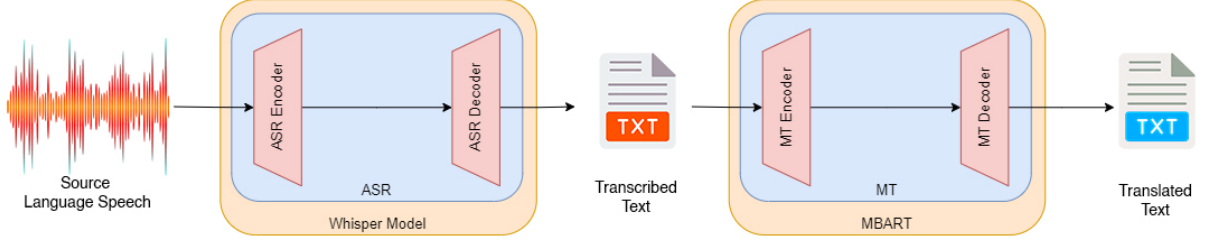


Figure 4: Overview of the cascade model for ST

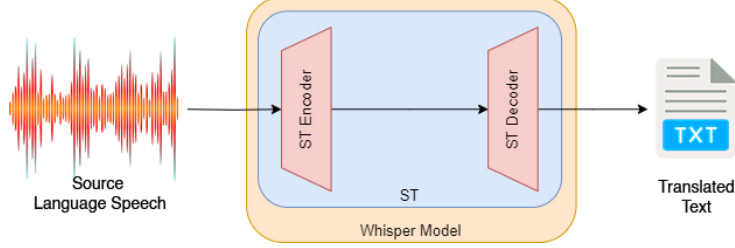


Figure 5: Overview of the end-to-end model for ST

Table 7: Speech translation results using cascaded (ASR + MT) methods on CoVoST-2 Es->En test sets in terms of BLEU score.

X ->En	Es
Cascaded ASR+MT Methods	
SeamlessM4T-Large (ASR+MT) [18, 19]	41.1
GenTranslate [18]	42.1
GenTranslate-V2 [18]	43.3
Whisper-Large-V2 + mBART-50 [18, 15]	40.7
Proposed Method + mBART-Large-50	31.13

Table 8: Speech translation results using end-to-end methods on CoVoST-2 Es->En test sets in terms of BLEU score.

X ->En	Es
End-to-end ST Methods	
SeamlessM4T-Large [18, 19]	41.1
GenTranslate [18]	42.0
GenTranslate-V2 [18]	43.6
Whisper-Large-V2 [18, 15]	39.6
Proposed Method	41.8

6.1 End-to-End Model

The end-to-end approach using Whisper-Large-V3 achieves a BLEU score of 41.8, making it one of the top-performing models in this category. This result demonstrates the capability of Whisper Large v3 to handle both automatic speech recognition (ASR) and machine translation (MT) tasks in a single, unified framework.

6.2 Cascade Model

In the cascade approach, the Whisper-Large-V3 model is paired with mBART-Large-50 for ASR and MT tasks. Interestingly, the BLEU score for this setup drops significantly to 31.13, indicating a substantial decrease in performance compared to its end-to-end counterpart. This reduction is likely due to error propagation between the ASR and MT stages: inaccuracies in transcription from Whisper-Large-V3 adversely affect the input quality for mBART-Large-50, leading to suboptimal translations.

7 Conclusion

This study compared both approaches using Whisper-Large-V3 for ASR and mBART-Large-50 for MT in the cascade setup, while the end-to-end system relied solely on Whisper-Large-V3. The results demonstrate that while cascade models can achieve high translation quality under ideal conditions, they are significantly affected by ASR errors. In contrast, end-to-end models provide a more robust and efficient solution, maintaining better performance even in challenging scenarios.

In the end-to-end approach, the Whisper-Large-V3 model for ST achieves a BLEU score of 41.8, demonstrating its capability to perform direct translation task while simplifying the translation pipeline and mitigating cumulative errors by directly processing audio into the target translation.

Although GenTranslate-V2 [18] achieves a BLEU score of 43.6, it still relies on a two-stage pipeline (generating N-best hypotheses from a foundation model followed by LLM-based integration). However, Whisper performs direct end-to-end translation within a single model, eliminating intermediate error accumulation.

Although GenTranslate-V2 achieves marginally higher BLEU scores by leveraging the reasoning capabilities of large language models, Whisper’s self-contained design offers greater efficiency and broader applicability in real-world scenarios where computational resources or LLM access may be limited.

In the cascade approach, the Whisper-Large-V3 for ASR model achieves a WER score of 9.36, indicating a relatively strong transcription performance, and the mBART-Large-50 model for MT achieves a BLEU score of 43.3 when provided with correctly transcribed text. When Whisper Large-V3 is paired with mBART-Large-50 in a cascade setup, the final BLEU score drops drastically to 31.13.

This significant decline suggests that transcription errors from the ASR stage propagate into the MT stage, reducing translation quality. In contrast, the end-to-end Whisper-Large-V3 model outperforms the cascade system by maintaining contextual consistency and reducing error accumulation. These findings underscore the advantages of end-to-end speech translation models for improving overall translation robustness and accuracy in real-world applications.

Authors’ contribution

The authors confirm contribution to the paper as follows: BD: Conceptualization, Investigation, Software, Data curation, Writing – Original draft; RF: Supervision, Validation, Writing – review & editing. All authors reviewed the results and approved the final version of the manuscript.

Competing interests

The authors have declared that no competing interests exist.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, we used ChatGPT, developed by OpenAI, to help search for new research on this topic and to improve the readability and language of some paragraphs of the manuscript. After using this tool, we reviewed and edited the entire content and assumed full responsibility for the content of the publication.

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